

Cybersecurity Guide

A Simple Reference on Core Concepts and Threats

1. What is Cybersecurity?

Cybersecurity is the practice of protecting systems, networks, programs, devices, and data from digital attacks, unauthorized access, or malicious damage. As technology becomes central to daily life and business, implementing strong security measures is critical to safeguard personal, financial, and operational information.

2. The Core Principles (The CIA Triad)

Information security is built around three foundational goals, often called the CIA Triad:

- **Confidentiality:** Making sure sensitive data is only accessed by people who are authorized to see it. This is typically managed through passwords, encryption, and access controls.
- **Integrity:** Making sure data remains accurate and unaltered. This ensures that files or messages are not changed or tampered with during transmission or storage.
- **Availability:** Making sure systems and data are reliable and accessible to authorized users whenever they need them. This involves keeping systems updated, maintaining backups, and preventing downtime.

3. Common Types of Malware

Malware (short for malicious software) is any code or program written specifically to harm, exploit, or steal from a computer system. Common types include:

- **Ransomware:** A type of malware that locks or encrypts your files so you can no longer access them. The hackers then demand a ransom payment in exchange for the decryption key.
- **Spyware:** Software that secretly installs itself on a device to monitor user activity. It can track keystrokes, log passwords, and steal personal details without the user's knowledge.
- **Trojan Horses:** Malicious programs that disguise themselves as legitimate software. Once a user downloads and opens them, they open a backdoor for attackers or deliver other harmful payloads.
- **Viruses and Worms:** Malware designed to replicate themselves. Viruses usually attach to a clean file and rely on human action to spread, whereas worms can spread autonomously across networks from computer to computer.

4. Other Common Attack Vectors

Hackers use various other methods to breach security defenses beyond just software exploits:

- **Phishing:** Deceptive emails, messages, or websites designed to mimic real companies (like banks or tech services). The goal is to trick users into typing in their passwords or credit card numbers.
- **Man-in-the-Middle (MitM) Attacks:** An attack where a hacker intercepts communication between two parties—such as your laptop and a public Wi-Fi network—to steal data as it travels over the air.
- **Denial of Service (DoS / DDoS):** An attempt to crash a website or online service by flooding it with an overwhelming amount of fake traffic, making it unavailable to legitimate users.

5. Basic Practices for Defense

Protecting a digital environment requires continuous, standard security habits:

- **Multi-Factor Authentication (MFA):** Requiring more than just a password to log in, such as a temporary text code or an authenticator app.
- **Regular Software Updates:** Keeping operating systems and applications updated fixes the software bugs and security holes that hackers rely on to break into systems.
- **Data Backups:** Keeping separate, secure copies of important files ensures that data can be recovered in the event of a ransomware attack or hardware failure.